

Experiment 4

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1 Aim

To analyse the γ -ray spectra of $^{137}_{55}\text{Cs}$ and $^{60}_{27}\text{Co}$ using Multi-Channel Analyzer (MCA). And observing the variation of resolution of photo-peaks in Co-60 spectra with increasing PMT voltage.

2 Experimental setup



Figure 1: Experimental setup of the instruments at the laboratory

In this experiment, we first set the PMT voltage to 610V, Gain to 8 and the LLD to 1 as mentioned on the apparatus. We then use the $^{60}_{27}\text{Co}$ source and take a reading for 10 minutes to calibrate the energy of the spectrum. This calibration involves the background elimination with a polynomial of order 2 and the Gaussian multipeak fitting for the 2 peaks to find centroids. This gives us a channel no. vs energy relation (i.e calibration of MCA). The energy of the photo peak of Cs-137 is then calculated. Then, we calculate the relative Photo-peak efficiency for Co-60 peaks using the ratio of total area of the spectrum to Photo-peak area. Finally, we proceed to analyse the variation of Photo-peak resolutions with increasing PMT voltage.

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3 Working Principle

3.1 Multi-Channel Analyzer (MCA)

An MCA works in the following steps:

- **Signal Generation:** When radiation (such as gamma rays) interacts with a detector, it generates an electrical pulse whose amplitude corresponds to the energy of the incident radiation.
- **Pulse Generation:** Now, when this radiation interacts with the detector, it generates an electrical pulse whose amplitude corresponds to the energy of the incident radiation.
- **Pulse Height Discrimination:** The incoming signal is processed through a pulse height discriminator (or LLD), which ensures that only signals above a certain threshold are considered. This eliminates noise and low-energy signals that are not of interest.
- **Analog-to-Digital Conversion:** The valid pulses are fed into an Analog-to-Digital Converter (ADC). The ADC samples the pulse height (voltage) and converts it into a digital value representing the energy of the pulse.
- **Channel Allocation:** The MCA divides the digital output into discrete channels, each representing a specific energy range. The number of channels can vary, but typical configurations might have several thousand channels.
- **Counting and Storage:** Each time a pulse is detected, it is assigned to the corresponding channel based on its energy. The MCA increments the count for that channel. Over time, this process builds up a histogram of counts versus energy levels.

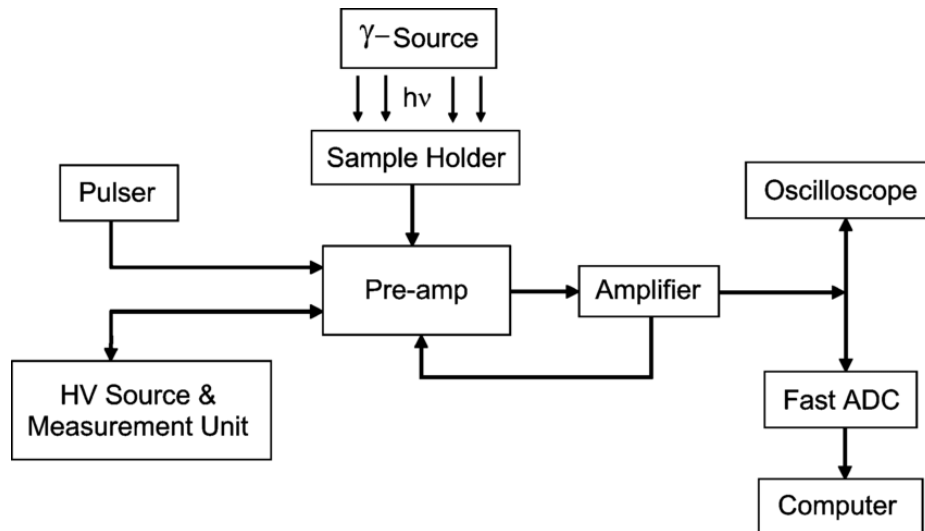


Figure 2: A schematic diagram of the working of a Multi-Channel Analyser. Source [Research Gate](#)

4 Theory

4.1 Decay processes of $^{137}_{55}\text{Cs}$ and $^{60}_{27}\text{Co}$

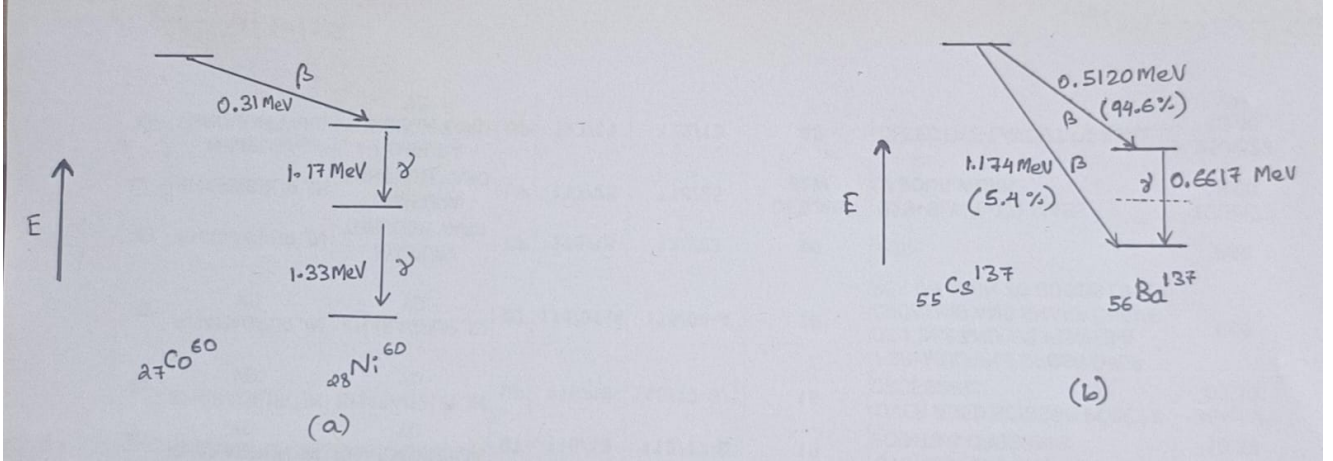


Figure 3: Shell model diagram of (a) $^{60}_{27}\text{Co}$ and $^{60}_{28}\text{Ni}$ and (b) $^{137}_{55}\text{Cs}$ and $^{137}_{56}\text{Ba}$ nuclei and their decay scheme.

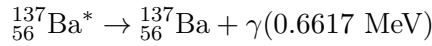
In this experiment we use the radio-nuclide of $^{137}_{55}\text{Cs}$ and $^{60}_{27}\text{Co}$. The decay processes occur in two steps. The detailed decay schemes of the nuclide are as follows:

Caesium-137 ($t_{1/2} \approx 30.1$ yrs)

$^{137}_{55}\text{Cs}$ decays into an excited $^{137}_{56}\text{Ba}$ nucleus, producing a β^- and an antineutrino.



Then the excited $^{137}_{56}\text{Ba}$ nucleus then decays into ground state $^{137}_{56}\text{Ba}$ nucleus, producing γ -ray in the process.

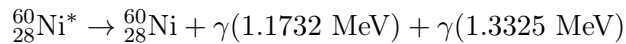


Cobalt-60 ($t_{1/2} \approx 5.27$ yrs)

$^{60}_{27}\text{Co}$ decays into an excited $^{60}_{28}\text{Ni}$ nucleus, producing a β^- and an antineutrino.



Then the excited $^{60}_{28}\text{Ni}$ nucleus then decays into ground state $^{60}_{28}\text{Ni}$ nucleus in a 2 step process, producing two γ -rays of different energies.



In the γ ray spectrum of these, we note some of the features that are observed-

Backscattering Peak

The fact that electrons often undergo large-angle deflections along their tracks leads to the phenomenon of backscattering. The backscattering peak results from gamma rays that are scattered in the surrounding materials (like walls, detector housing, etc.) and then detected. This peak typically appears at lower energies, around 150-250 keV, depending on the geometry and materials involved.

Compton Continuum

The Compton continuum arises from the Compton scattering process, where a gamma photon is partially absorbed, and the rest of the energy is scattered. The scattered photon escapes, leaving behind a range of energies in the detector.

Photo-peak

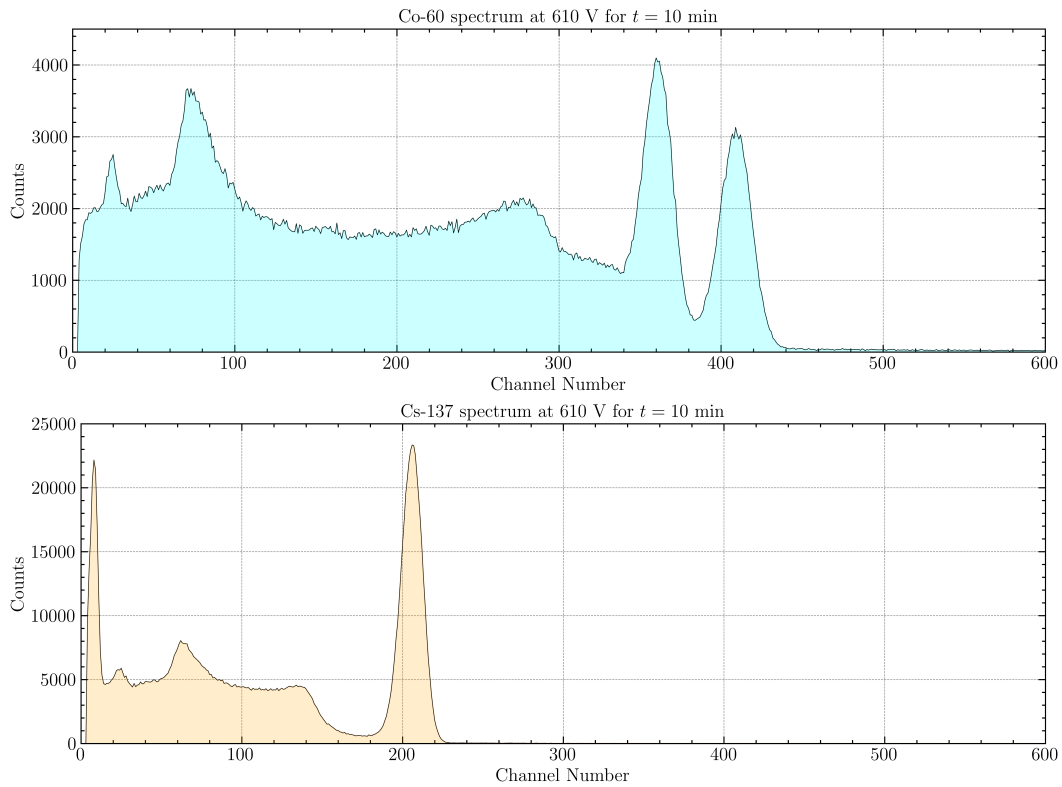
The Photo-peak represents the full-energy peak and occurs when the gamma ray transfers all its energy to an electron in the detector through the photoelectric effect. $^{60}_{27}\text{Co}$, emits two primary gamma rays at 1.17 MeV and 1.33 MeV, (corresponding to its two major transitions). So would see *two photo-peaks* in its spectrum. On the other hand, $^{137}_{55}\text{Cs}$ emits a gamma ray at 0.6617 MeV (corresponding to its only major transition, after a β decay), thus we will see a single Photo-peak corresponding to this energy.

Single and Double Escape Peaks ($^{60}_{27}\text{Co}$)

For higher energy gamma rays, like those from $^{60}_{27}\text{Co}$, the spectrum may show escape peaks. These occur when an electron-positron pair is produced, and one or both annihilation photons (511 KeV each) escape the detector.

5 Data Analysis

We performed our calibration, against the known energies of photo-peaks in Co-60 spectra, and then utilized these calibration constants to determine the energy of Photo-peak in Cs-137 spectra. Both of which were taken at a PMT voltage of 610 V and for $t = 10$ mins.



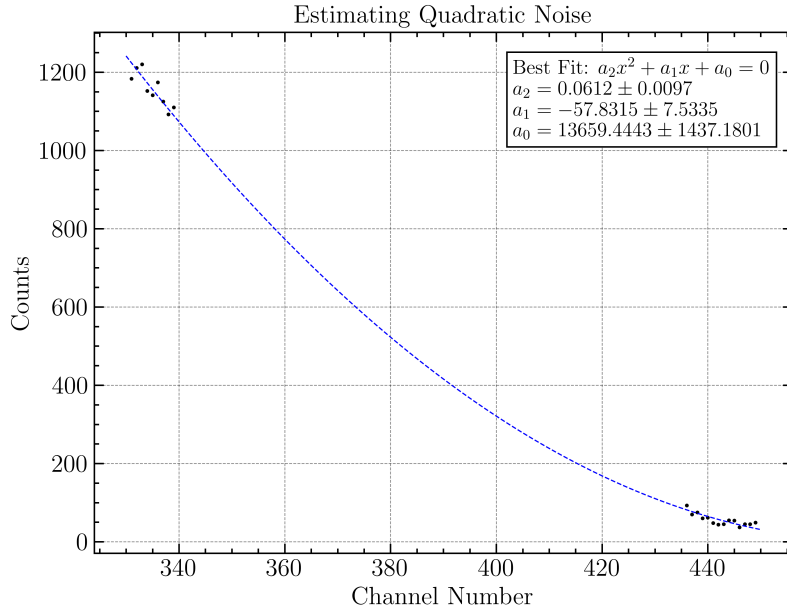
5.1 MCA Calibration

Estimating Non-Linear background

The two photo-peaks in Co-60, need to be Gaussian fitted to find the centroid channel numbers of the corresponding peaks. But in order to do so we need to adjust their baselines. We achieve this by first estimating a non-linear background. We implemented a quadratic noise model for this.

We have

$$f_n(x) = a_2x^2 + a_1x + a_0 = 0$$



where,

$$\begin{aligned} a_0 &= 13659.4443 \pm 1437.1801 \\ a_1 &= -57.8315 \pm 7.5335 \\ a_2 &= 0.0612 \pm 0.0097 \end{aligned}$$

Multi-peak fitting of Co-60 spectra

We subtract this quadratic noise from our counts data to get the baseline adjusted (`scaled_counts`). We proceed to fit this data, with a sum of 2 gaussians and an offset ($f_G(x)$)-

$$\begin{aligned} f_G(x) &= A_0 + N(\mu_1, \sigma_1) + N(\mu_2, \sigma_2) \\ &= A_0 + A_1 \exp\left\{-\frac{(x - \mu_1)^2}{2\sigma_1^2}\right\} + A_2 \exp\left\{-\frac{(x - \mu_2)^2}{2\sigma_2^2}\right\} \end{aligned}$$

The results of the fitting (see Fig 4) are summarised below

$$\begin{aligned} A_0 &= -56.06754 \pm 2.7714 \\ A_1 &= 3342.3894 \pm 23.3021 \\ \mu_1 &= 360.9855 \pm 0.0653 \\ \sigma_1 &= 8.1169 \pm 0.0654 \\ A_2 &= 2870.9954 \pm 21.9932 \\ \mu_2 &= 409.5029 \pm 0.0806 \\ \sigma_2 &= 9.1120 \pm 0.0806 \end{aligned}$$

Calibration of channels

We have, the γ -ray photon energy to be linearly related to the channel numbers. Let,

$$E_1 = \kappa\mu_1 + \epsilon$$

$$E_2 = \kappa\mu_2 + \epsilon$$

We know, $E_1 = 1.17$ MeV and $E_2 = 1.33$ MeV. Solving the system of equations, give

$$\begin{aligned} \kappa &= 0.0032977 \\ \epsilon &= -0.0204530 \end{aligned}$$

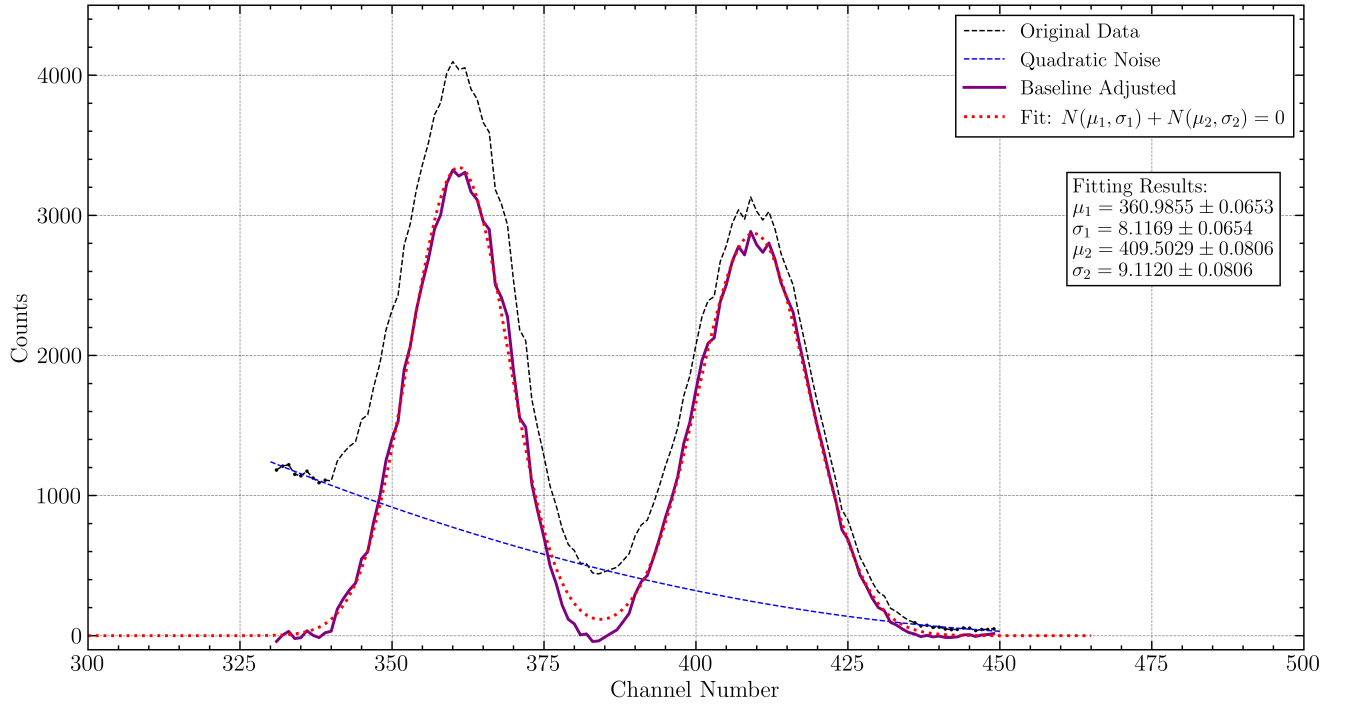


Figure 4: Multi-peak fitting with 2 gaussians on the baseline adjusted photo-peaks spectra of Co-60

5.2 Energy of Cs-137 peak

First we similarly find the centroid of the peak through gaussian fitting.

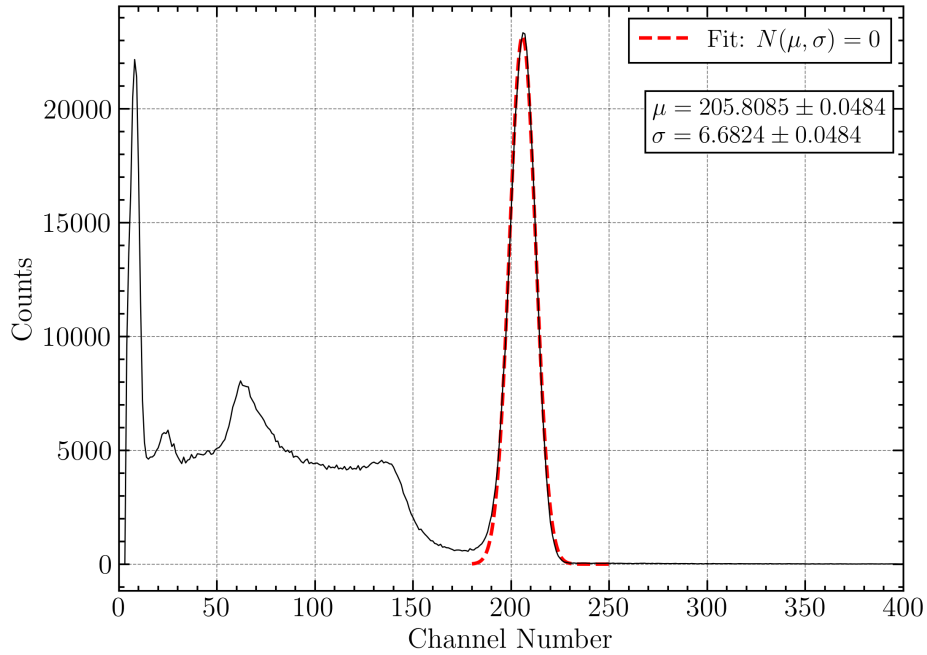


Figure 5: Gaussian fitting to find the centroid of the Photo-peak for Cs-137

Thus the energy corresponding to the Photo-peak (Fig 5) is calculated as

$$E_{Cs} = \kappa \mu_{Cs} + \epsilon$$

Substituting $\mu_{Cs} = 205.8085$, we get $E_{Cs} = 0.67265$ MeV.

5.3 Photo-peak Efficiency (for Co-60)

Photo-peak efficiency(η) is defined as the ratio of the area under the peak to that of the area of the whole spectrum.

$$\eta = \frac{\text{Area under peak}}{\text{Area under the whole spectrum}}$$

	1.17 MeV peak	1.33 MeV peak	Total
Area Counts	110649.0	78723.0	834825.0
η	0.1325	0.0943	

5.4 Variation of resolution for Co-60

The resolution is defined as

$$r = \frac{\Delta E}{E} = \frac{\text{FWHM of peak}}{\text{Centroid of peak}}$$

One must keep in mind that, the energy corresponding to each Photo-peak (E_i) is fixed for all PMT voltages. That is, the calibration constants need to be determined for each PMT voltage. However, we only need to work with resolution, which can be obtained by multi-peak fitting of all the individual spectra. The ΔE_i in each case is determined as

$$\Delta E_i = E_i \times \frac{\text{FWHM of peak}}{\text{Centroid of peak}}$$

From the literature (Krane, 1988) we take $E_1 = 1.1732$ MeV and $E_2 = 1.3325$ MeV. And for a gaussian distribution ($N(\mu, \sigma^2)$), the relation between FWHM and standard deviation is,

$$\text{FWHM} = \sigma \times 2\sqrt{2 \ln 2}$$

PMT Voltage (V)	E_1 (MeV)	ΔE_1 (MeV)	Resolution (r_1)	E_2 (MeV)	ΔE_2 (MeV)	Resolution (r_2)
560	1.1732	0.0560	0.0477	1.3325	0.0604	0.0453
570	1.1732	0.0570	0.0486	1.3325	0.0625	0.0469
580	1.1732	0.0551	0.0470	1.3325	0.0597	0.0448
590	1.1732	0.0532	0.0453	1.3325	0.0557	0.0418
600	1.1732	0.0583	0.0497	1.3325	0.0628	0.0471
610	1.1732	0.0549	0.0468	1.3325	0.0609	0.0457
620	1.1732	0.0547	0.0467	1.3325	0.0593	0.0445
630	1.1732	0.0624	0.0532	1.3325	0.0703	0.0527
640	1.1732	0.0625	0.0532	1.3325	0.0711	0.0533
650	1.1732	0.0617	0.0526	1.3325	0.0687	0.0515
660	1.1732	0.0604	0.0515	1.3325	0.0672	0.0504

We compare the resolution of the photo-peaks by varying the PMT voltage in Fig 6.

NOTE: Since resolving power $\propto \frac{1}{r}$, The optimum HV which is better able to resolve the photopeaks is the one which has the *minimum resolution* (not the maximum) for both of them.

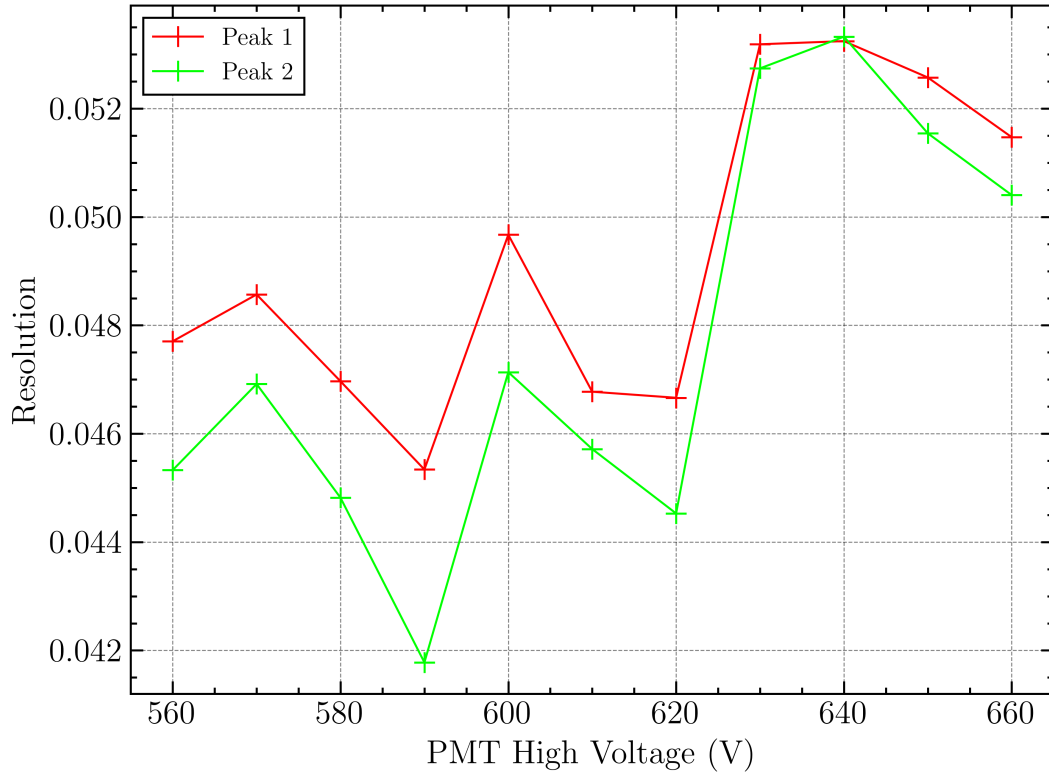


Figure 6: Plot of variation of resolution with increasing PMT voltage

6 Error Analysis

6.1 Error in Calibration Constants

We have,

$$\kappa = \frac{E_1 - E_2}{\mu_1 - \mu_2}$$

Taking log both sides and differentiating, thus finding the maximum possible error, we get

$$\begin{aligned} \left| \frac{\Delta\kappa}{\kappa} \right| &\leq \frac{|\Delta\mu_1 - \Delta\mu_2|}{|\mu_1 - \mu_2|} \leq \frac{|\Delta\mu_1| + |\Delta\mu_2|}{|\mu_1 - \mu_2|} \\ \Rightarrow |\Delta\kappa| &= 0.0000099167 \end{aligned}$$

Hence,

$$\boxed{\kappa = (3.2977 \pm 0.0099) \times 10^{-3}}$$

Further,

$$\begin{aligned} \epsilon &= E_1 - \kappa\mu_1 \\ \Rightarrow |\Delta\epsilon| &\leq |\kappa\Delta\mu_1| + |\mu_1\Delta\kappa| = 0.0020693869 \end{aligned}$$

Calculating

$$\boxed{\epsilon = (-2.0453 \pm 0.2069) \times 10^{-2}}$$

6.2 Error in energy of Cs peak

$$\boxed{E_{Cs} = \kappa\mu_{Cs} + \epsilon}$$

Similarly, maximizing the error

$$|\Delta E_{Cs}| \leq |\Delta\epsilon| + |\kappa\Delta\mu_{Cs}| + |\mu_{Cs}\Delta\kappa|$$

Substituting $\Delta\mu_{Cs} = \pm 0.0484$, we have

$$\boxed{E_{Cs} = (0.6726 \pm 0.0034) \text{ MeV}}$$

6.3 Sources of Error

Environmental Radiation Background radiation from the environment, cosmic rays, or other radioactive sources in the surroundings may cause noise and interfere with the gamma-ray peaks. We took a reading without any source and indeed found a reasonable amount of background noise. Since the distribution of background noise is not constant, the error could not be eliminated by simply subtracting the noise reading from the data.

Fitting Inaccuracy There is some inaccuracy in the data of resolutions obtained as the fitting is not perfect and the choice of initial parameter values affects the fit to a certain extent. The fitting can be made better by using Maximum Likelihood Estimation of the parameters.

7 Conclusion

The experiment provided a clear understanding of the operation of a scintillation detector and the interaction of gamma radiation with matter. During the experiment, we observed the double Photo-peak for cobalt and the single Photo-peak for cesium. The double Photo-peak of cobalt was utilised as a reference to calibrate and determine the energy of the single Photo-peak of cesium using *energy* versus *channel number* relation. After background correction using a quadratic polynomial, to adjust the baselines of the photo-peaks, the spectra was fitted using a sum of 2 gaussians. The key conclusions from the experiment are as follows:

- The energy of the single Photo-peak of cesium was determined to be (0.6726 ± 0.0034) MeV.
- We computed the photo-peak efficiency, obtaining values of 0.1325 and 0.0943 for the two photo-peaks of cobalt at energy levels of 1.17 MeV and 1.33 MeV, respectively.
- A graph of resolution versus voltage was plotted, showing no consistent trend, however the minimum resolution of both the peaks were obtained at $HV = 590$ V, suggesting it to be the optimum PMT voltage to resolve both the photopeaks prominently for this source.
- Furthermore, we observed that the peaks shifted to the right as the PMT voltage increased, which is consistent with the expected behaviour.

References

- Knoll, Glenn F., 2010 *Radiation detection and measurement*. John Wiley & Sons.
- Krane, K.S., 1988. *Introductory nuclear physics*. John Wiley & Sons.

Appendix

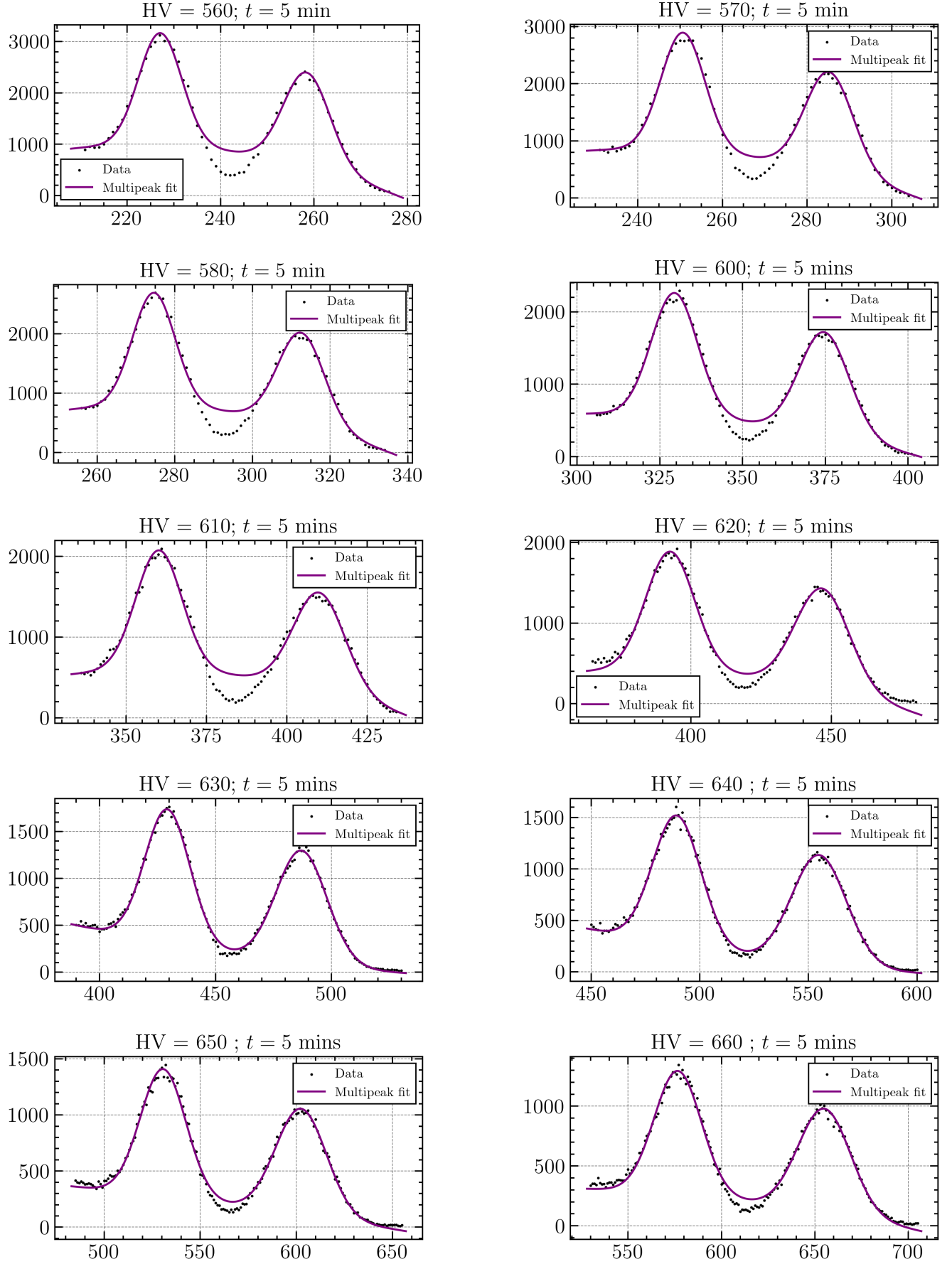


Figure 7: Fitted Photo-peak profiles with quadratic noise, of Co-60 spectra with increasing PMT voltage (at a step of 10V).